

Data Preparation and Business Questions

Date	12 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns

Data Preparation

The India Electricity Consumption dataset was collected in CSV format and prepared for analysis using Microsoft SQL Server. The dataset was examined for missing values, duplicate records, and inconsistent entries before importing it into the database. SQL queries were used to clean, transform, and organize the data for efficient analysis. The processed data was then connected to Tableau Public to build interactive dashboards and stories.

Data Preparation Steps

- Collected the India Electricity Consumption dataset.
- Removed duplicate and missing records.
- Standardized column names and data types.
- Imported the cleaned dataset into Microsoft SQL Server.
- Executed SQL queries for filtering and aggregation.
- Connected SQL Server to Tableau Public.
- Created dashboards and stories for visualization.

Business Questions

The project addresses the following business questions:

1. Which Indian states recorded the highest electricity consumption?
2. How did electricity consumption change between 2019 and 2020?
3. What was the impact of the COVID-19 lockdown on electricity usage?
4. Which region consumed the highest amount of electricity?
5. How does electricity consumption vary across months and quarters?
6. Which states showed the highest and lowest electricity consumption?
7. What insights can support better energy planning and resource allocation?